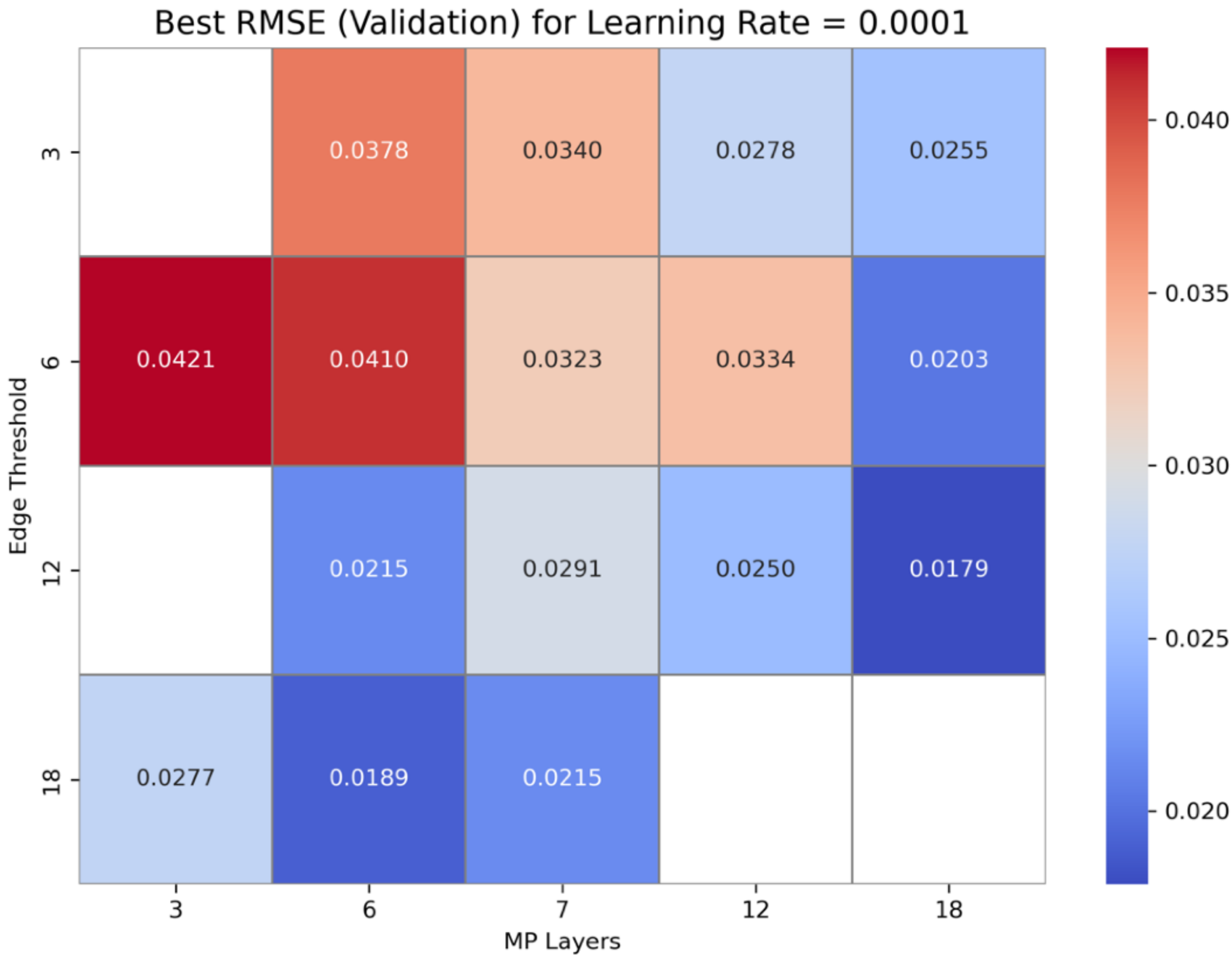
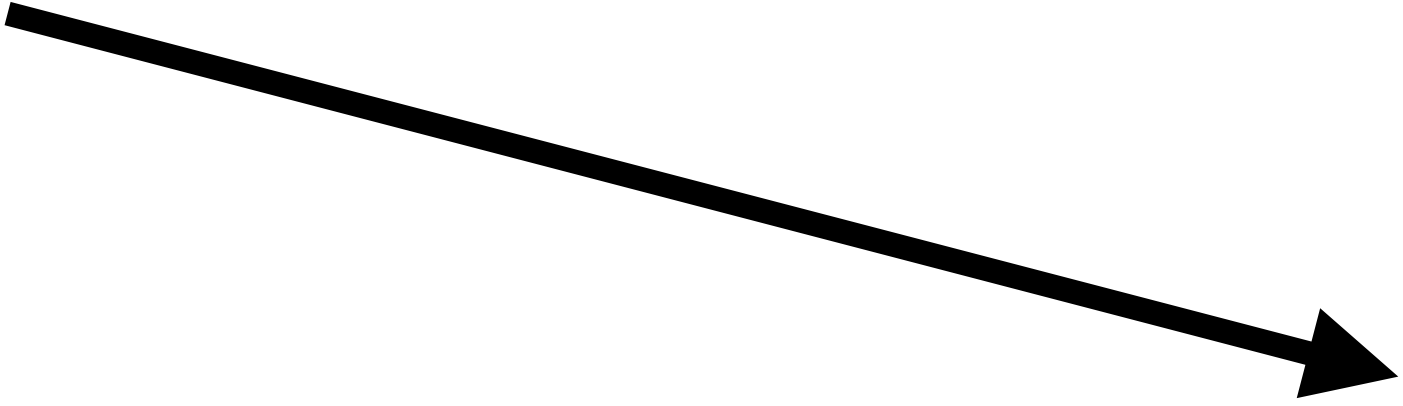
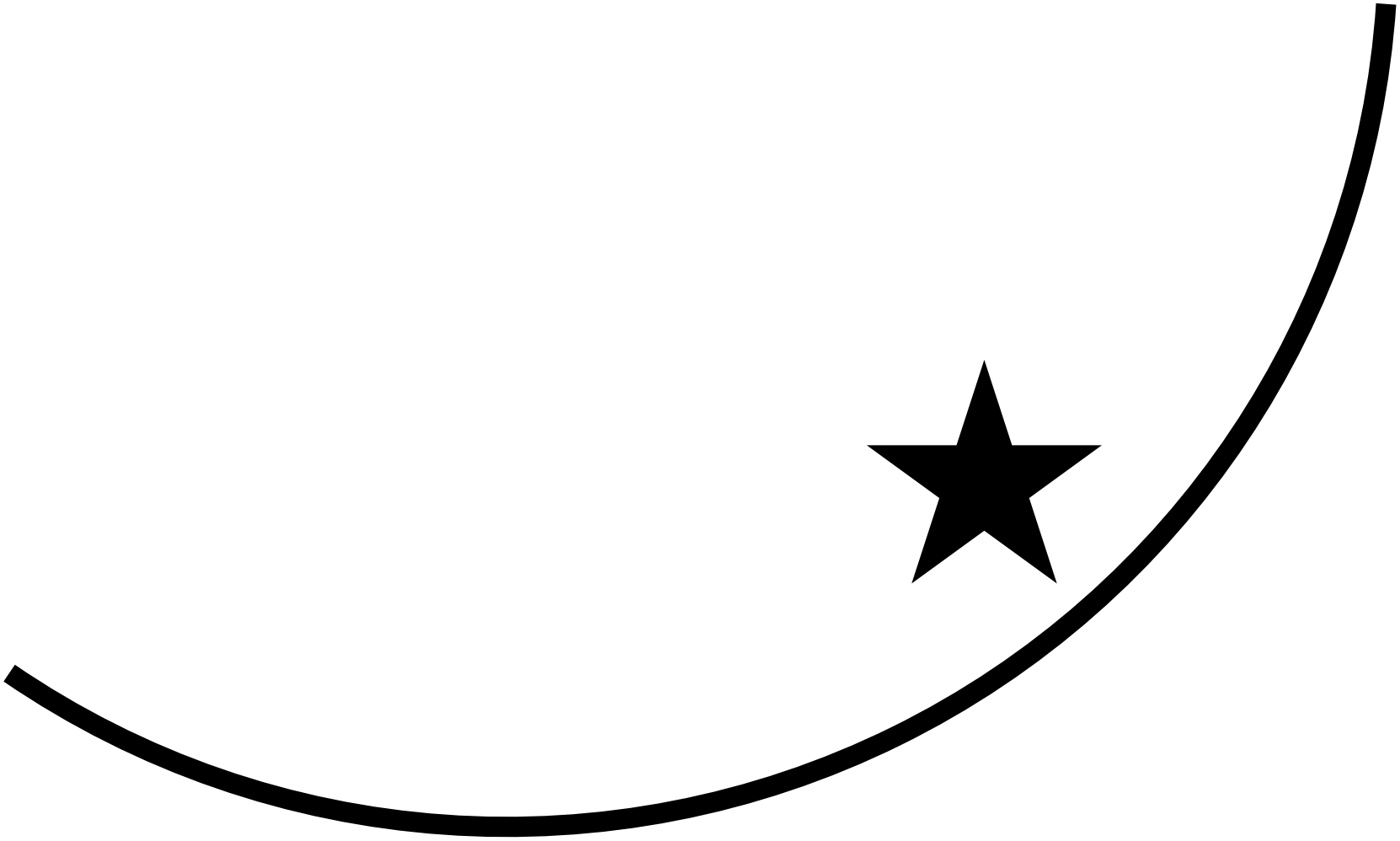
Tuning graph architecture

- After testing on ‘toy’ tasks, we focus on tuning number of message passing layers (MPL) and edge threshold (ET) — clarify edge threshold as distance in box


Best RMSE (Validation) for Learning Rate = 0.0001
Edge Threshold
MP Layers
3
6
12
18
3
6
7
12
18
0.0378
0.0340
0.0278
0.0255
0.0421
0.0410
0.0323
0.0334
0.0203
0.0215
0.0291
0.0250
0.0179
0.0277
0.0189
0.0215
0.040
0.035
0.030
0.025
0.020





21

- Large models were performant but cumbersome, erratic, and divorced from physics

- Our choice: 10 (MPL) and 6 (ET)

Tuning graph architecture

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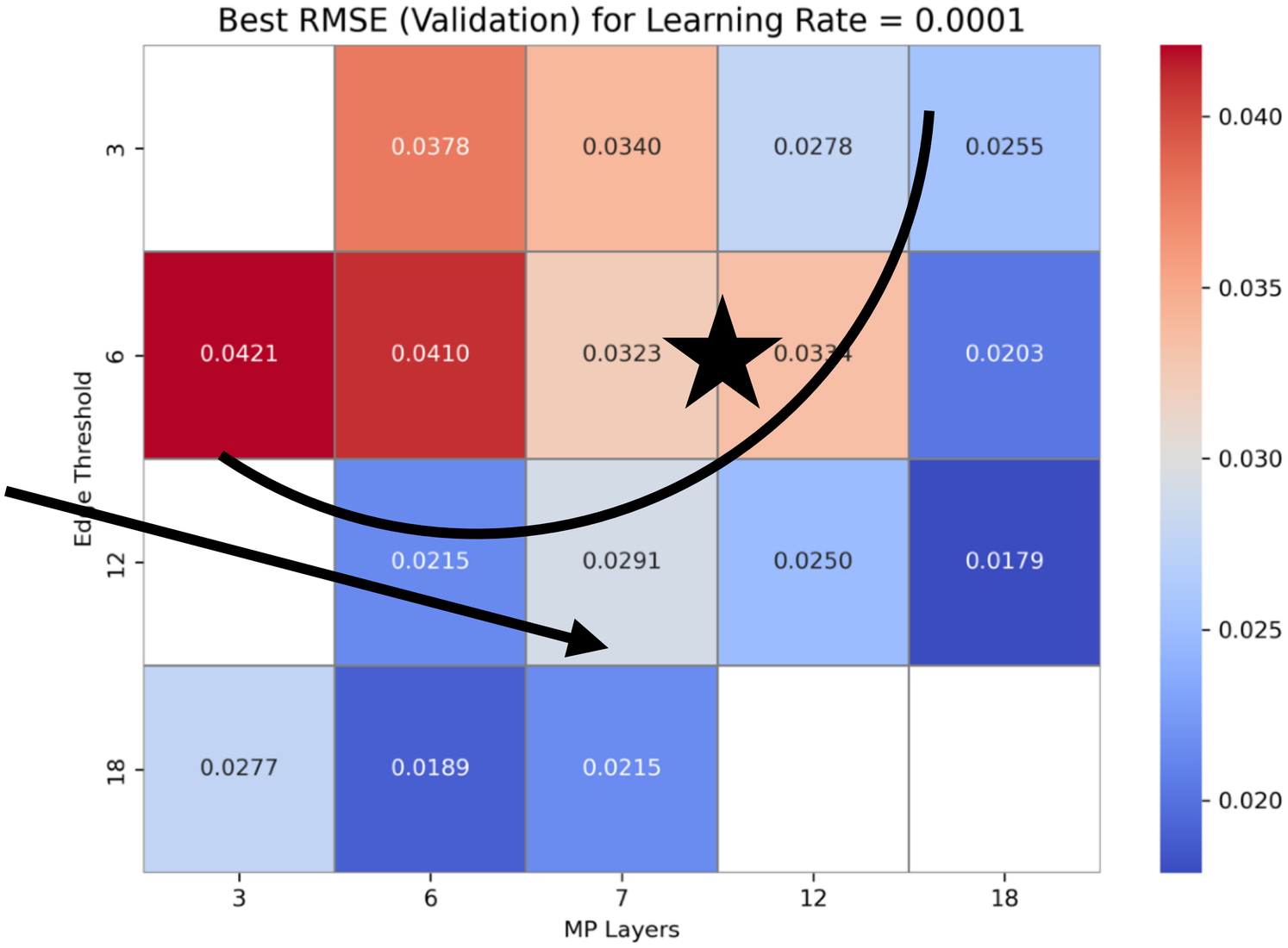
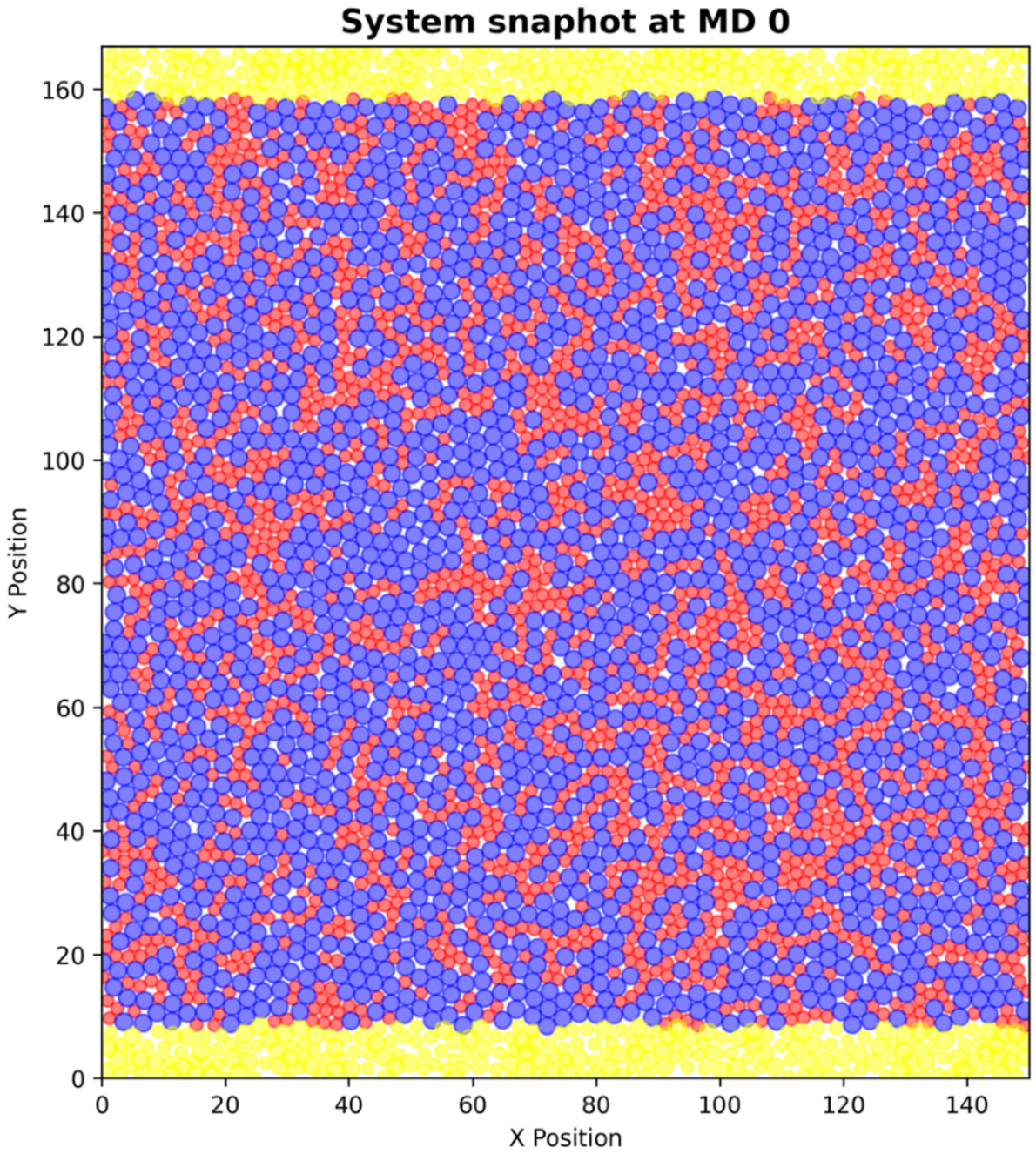
The physics of our parameter choices